U.S. Particle Physics

Leadership, Innovation, and Discovery




Fermilab

The U.S. particle physics community asks for your support of the P5 Report's strategic plan by providing FY2027 appropriations that include:

$1.385B for High Energy Physics including at least $360M (26%) for core research, within a budget of $9.5B for the Department of Energy's Office of Science, and $9.9B for the National Science Foundation.

This level of funding will enable the U.S. to drive world-leading high energy physics research, advance the highest priority large-scale projects, sustain facility operations, and promote a portfolio of small and medium-sized projects. Specifically, core research funding provides critical support for scientific researchers and students at universities and national laboratories to advance experimental and theoretical research, conduct accelerator and detector R&D, and train the next-generation STEM workforce. Investment in high energy physics also underpins U.S. leadership in Quantum Information Science, Artificial Intelligence, and Microelectronics; where innovations from our field strengthen national competitiveness, technological leadership, and economic security.

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Specifically, this level of funding is needed to advance the major priorities of the P5 strategic plan by:

1. **Restoring and growing core research** at universities and national laboratories to maintain U.S. leadership and advance groundbreaking scientific discoveries;

2. **Ramping up construction of world-class science facilities** and state-of-the-art instrumentation, including P5's highest priorities—accelerator and detector upgrades at the Large Hadron Collider (LHC), the Long-Baseline Neutrino Facility (LBNF) and the Deep Underground Neutrino Experiment (DUNE);

3. **Funding operations of experiments and facilities** to support the scientific research of thousands of users from national laboratories and academic institutions;

4. **Building the future of high energy physics through focused R&D** in accelerators, instrumentation, computing, and theory, as well as cutting-edge research and innovation in emerging technology areas including Quantum Information Science, Artificial Intelligence/Machine Learning, and Microelectronics; and

5. **Expanding workforce and education programs in STEM** to attract and invest in the next generation of top domestic and international talent.

Physics Propels U.S. Progress

Particle physicists endeavor to *understand matter, energy, space, and time*—from the *tiniest particles* to the *entire cosmos*. Investment in fundamental research enables the U.S. to champion global collaboration, spearhead scientific breakthroughs, and power innovation.



Exploring the Quantum Universe

America's particle physics research program transforms our understanding of the universe by pursuing some of humankind's most profound questions:

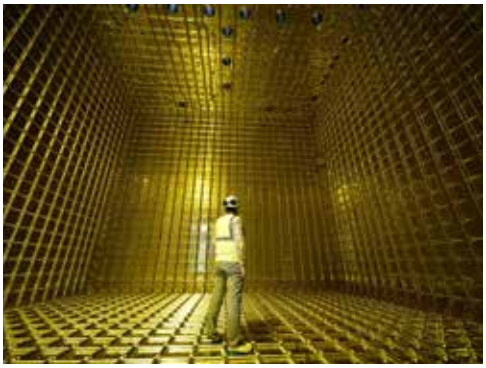
- How did the **universe evolve** into its present form?
- What are the **fundamental building blocks of matter**?

These questions inspire the construction of world-leading facilities, from particle accelerators to telescopes, to illuminate the sweeping connections between nature's smallest and largest structures.

Investing in the Future

Sustained progress in particle physics over the next decade requires continued and amplified investment to:

- Train the **next generation of scientists** and build a **competitive workforce** to steer technological advancement
- Construct critical **technological** and **computational infrastructure** to ensure U.S. leadership in cutting-edge research
- Advance **emerging technologies** and **theoretical frameworks** to facilitate cross-disciplinary synergies with industry





Driving Advancements in Technology R&D

The U.S. particle physics community comprises a network of scientists, engineers, technicians, and students whose work continues to galvanize innovation across a wide range of sectors:

- **Medicine**: Improve diagnostics & treatment of disease
- **Security**: Sharpen nuclear monitoring & cyberinfrastructure
- **Energy**: Strengthen breakthroughs in nuclear fusion
- **Computing**: Enhance quantum systems & artificial intelligence
- **Industry**: Promote manufacturing & microelectronics partnerships
- **Geology + Archeology**: Study natural & manmade structures

The U.S. particle physics community's vision is guided by intertwined science drivers—*Deciphering the Quantum Realm*, *Illuminating the Hidden Universe*, and *Exploring New Paradigms of Physics*—that represent the most promising avenues of investigation.

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Progress & Priorities

U.S. particle physics advances the frontiers of discovery science, builds and trains a highly skilled technical workforce, and promotes U.S. leadership in a worldwide context.

P5 provides the long-term strategy and priorities for U.S. investments in particle physics for the next 10 years within a 20-year vision. The 2023 P5 report reaffirms the ongoing priorities and outlines future opportunities for discovery.

The top priorities in 2026 are to advance current construction projects, support operations of ongoing experiments, and strengthen support for research at universities and national laboratories to maximize the opportunities for scientific discoveries.

Support for big data analysis, innovative instrumentation development, a vibrant theory program, and next-generation accelerator technology is essential to maintain U.S. leadership and to train the next generation of scientists and innovators. U.S. particle physics also plays a key role in advancing and leveraging national initiatives in microelectronics, artificial intelligence and machine learning (AI/ML), and quantum information science (QIS).

It is critical to complete construction and deliver science from the following projects:

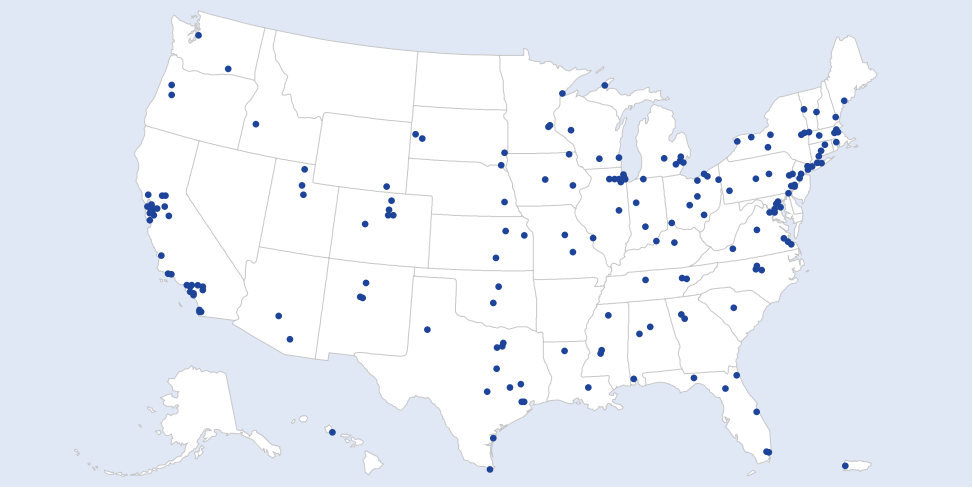
High-Luminosity Large Hadron Collider (HL-LHC) is an accelerator and detector upgrade for the CMS and ATLAS experiments to continue the highly successful LHC program and bilateral partnership with CERN. This project will enable new precise measurements of the Higgs boson, including discovery of its unique property of self-interaction. Searches for beyond the Standard Model physics using state-of-the-art machine learning techniques at the highest center of mass energy ever achieved in a collider experiment will broaden our understanding of fundamental particles and interactions.

The **Long-Baseline Neutrino Facility / Deep Underground Neutrino Experiment (LBNF/DUNE)** is the first U.S.-hosted world-leading international science mega-project. It will determine the fundamental properties of neutrinos, which are mysterious particles that permeate the cosmos. A new beam of neutrinos will be powered by a superconducting linear accelerator under construction by the PIP-II project at Fermilab in Illinois and delivered 800 miles through the earth to LBNF and into the neutrino detectors of DUNE, co-located at the Sanford Underground Research Facility in South Dakota.

These carefully chosen investments will enable a steady stream of exciting new results for many years to come and will maintain U.S. leadership in key areas.

Particle Physics in the United States

Particle physics is both global and local. Scientists, engineers, and technicians at over 200 universities, institutes, and laboratories throughout the U.S. are working in partnership with their international colleagues to build high-tech tools and components and conduct scientific research. Particle physics excels at building a highly-skilled technical workforce across the country by training and educating the next generation of innovators. Particle physics activities in the U.S. attract leading scientists from



The P5 strategy continues to be very successful. The 2023 P5 report builds upon these successes and charts the course to a new era of discovery, innovation, and U.S. leadership for the next decade and beyond.

Recent Science Results

NSF-DOE Vera C. Rubin Observatory has completed the construction of its major new telescope in Chile and installed the LSSTCamera. The Rubin Observatory has released its first images, marking a significant milestone in its commissioning. These initial observations confirm the exceptional capabilities of the LSSTCamera and the observatory's readiness to commence its groundbreaking Legacy Survey of Space and Time (LSST), which will provide unprecedented insights into cosmic evolution and astrophysical phenomena.

The **Dark Energy Spectroscopic Instrument** (DESI) presented results from its third year of data collection, releasing the largest 3D map of the universe to date. These findings provide enhanced constraints on dark energy and further hint that the rate of expansion of the cosmos has varied over time, further solidifying DESI's role in understanding fundamental cosmology.

The **LUX-ZEPLIN** (LZ) experiment has set a world-leading constraint in the search for galactic dark matter. Its latest results significantly refine the understanding of dark matter's properties, providing the most sensitive direct detection limits to date and further advancing the global effort to uncover this elusive component of our universe.

The **CMS** experiment at CERN has observed evidence for the existence of quasi-bound states of top quarks, known as toponium. The **ATLAS** experiment has made significant strides in understanding the fundamental properties of the Higgs boson by studying its longitudinal polarization. These investigations provide crucial insights into the Standard Model with unprecedented precision and a unique window to study these particle interactions at the highest energy scales.

The Breakthrough Prize in Fundamental Physics was awarded to all the Large Hadron Collider (LHC) collaborations: ATLAS, CMS, ALICE and LHCb, for testing the Standard Model and other theories describing physics that might lie beyond it to high precision. This recognition highlights the LHC's profound impact on understanding the nature of the universe.

The **Muon g-2 experiment** at Fermilab has delivered its final, most precise measurement of the muon's anomalous magnetic moment. This result reduces the experimental uncertainty to 1 part in 10 million, providing the most stringent test yet of the Standard Model's predictions for this fundamental particle property.

The **MicroBooNE experiment** has completed its search for evidence of a sterile neutrino, a hypothesized particle that could explain anomalies observed in previous experiments. These results are consistent with Standard Model predictions, significantly constraining the existence of such a particle and providing valuable insights for future neutrino research.

The **NOvA** and **T2K experiments** have presented a combined analysis of neutrino oscillations, yielding the most precise determination of key parameters governing neutrino behavior. This joint effort significantly advances our understanding of the neutrino mass ordering and the potential for charge-parity violation in the lepton sector, crucial for explaining the matter-antimatter asymmetry in the universe.

Program Advances and Milestones

Particle physicists have decades of experience pioneering and developing AI/ML techniques. Globally, collaborations continue to refine these techniques to better understand and exploit the largest scientific data sets in the world. These efforts will only become more important in the future of particle physics.

The Short-Baseline Near Detector (**SBND**) at **Fermilab** officially started collecting physics-quality neutrino data. As the final component of the SBN program, it is already recording the largest dataset of neutrino-argon interactions to date to measure the neutrino beam flux.

The Super Cryogenic Dark Matter Search (**SuperCDMS**) experiment successfully cooled their detectors to nearly absolute zero, enabling the search for light dark matter with unprecedented sensitivity. This achievement positions the collaboration to begin its first science run, potentially yielding groundbreaking discoveries in particle physics.

The five National Quantum Information Science Research Centers (**NQISRCs**) have deepened the scientific understanding of the physics underpinning quantum technologies. The centers have played a crucial role in training the next-generation quantum workforce and fostering collaborations across national laboratories, universities, and tech companies.

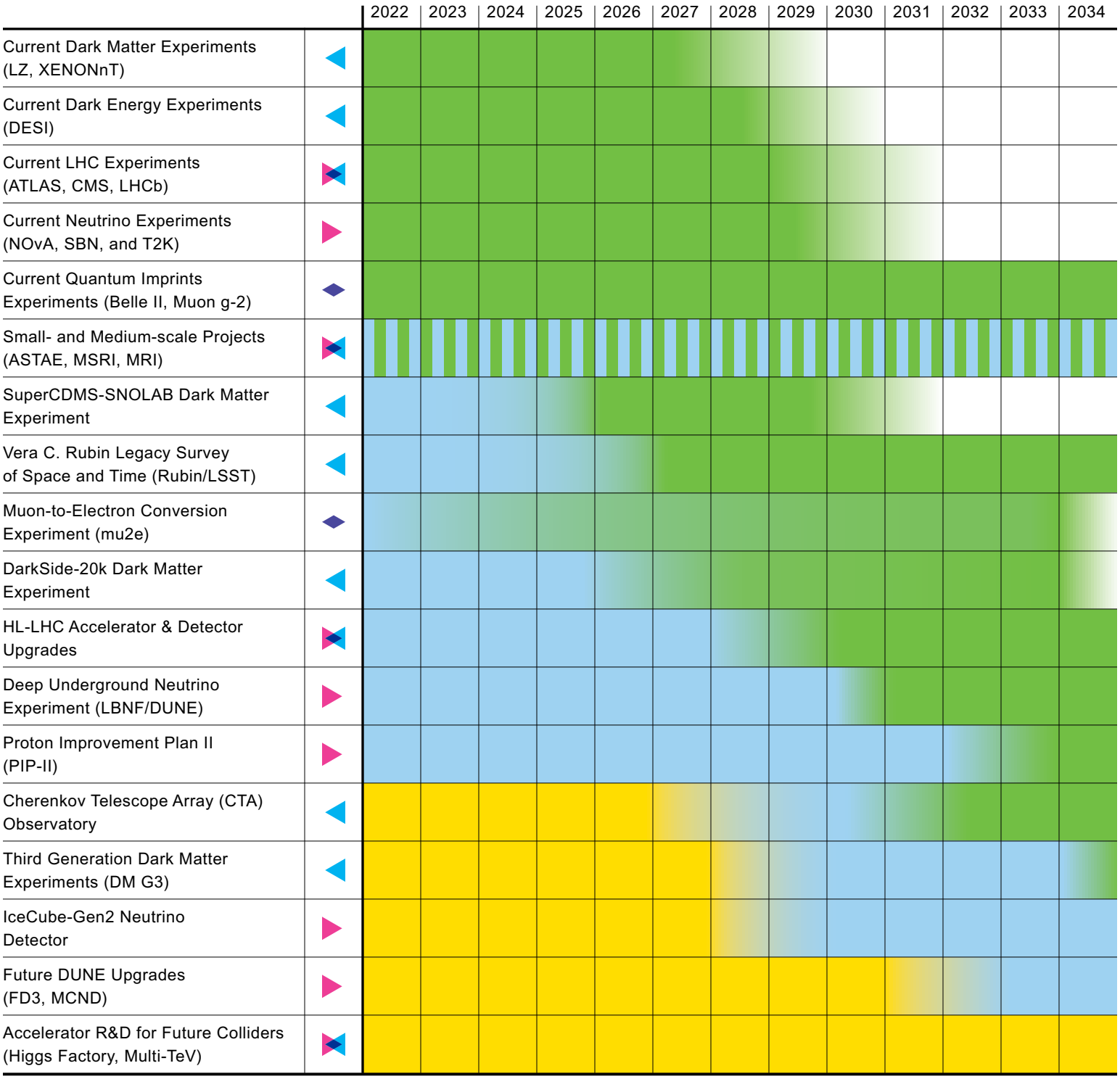
Theoretical physicists have gained insight into quantum effects involving black holes and branching of separate universes, and connections to quantum information. Lattice Quantum Chromodynamics (QCD) has become an indispensable tool in theoretical physics, particularly in refining our understanding of the muon's anomalous magnetic moment (g-2) and in determining parton distribution functions (PDFs).

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Particle Physics Experiments Timeline

Phase of Experiment
Operation & Analysis
Fabrication/Construction
Conceptual & Technical Design

Science Themes	Science Drivers
Decipher the Quantum Realm	Neutrinos, Higgs Boson
Illuminate the Invisible Universe	Dark Matter, Cosmic Evolution
Explore New Paradigms in Physics	New Particles, New Phenomena





Particle Physicists

Build STEM Leaders

Our public engagement programs, reaching tens of thousands of people every year, promote interest in STEM careers and demonstrate how science makes a difference in our lives.





Engaging our community

*"**SAGE summer camps** introduce [high schoolers] to the work and daily life of US National Laboratories scientists and engineers."*

Giulia Lanza, SLAC National Accelerator Laboratory

*"**The Big Bang Science Fair** caters to ages 3 to 99."*

Meenakshi Narain, Brown University

*"As part of the **QuarkNet program**, we invite high school students to do a crash-course in particle physics."*

Joe Haley, Oklahoma State University

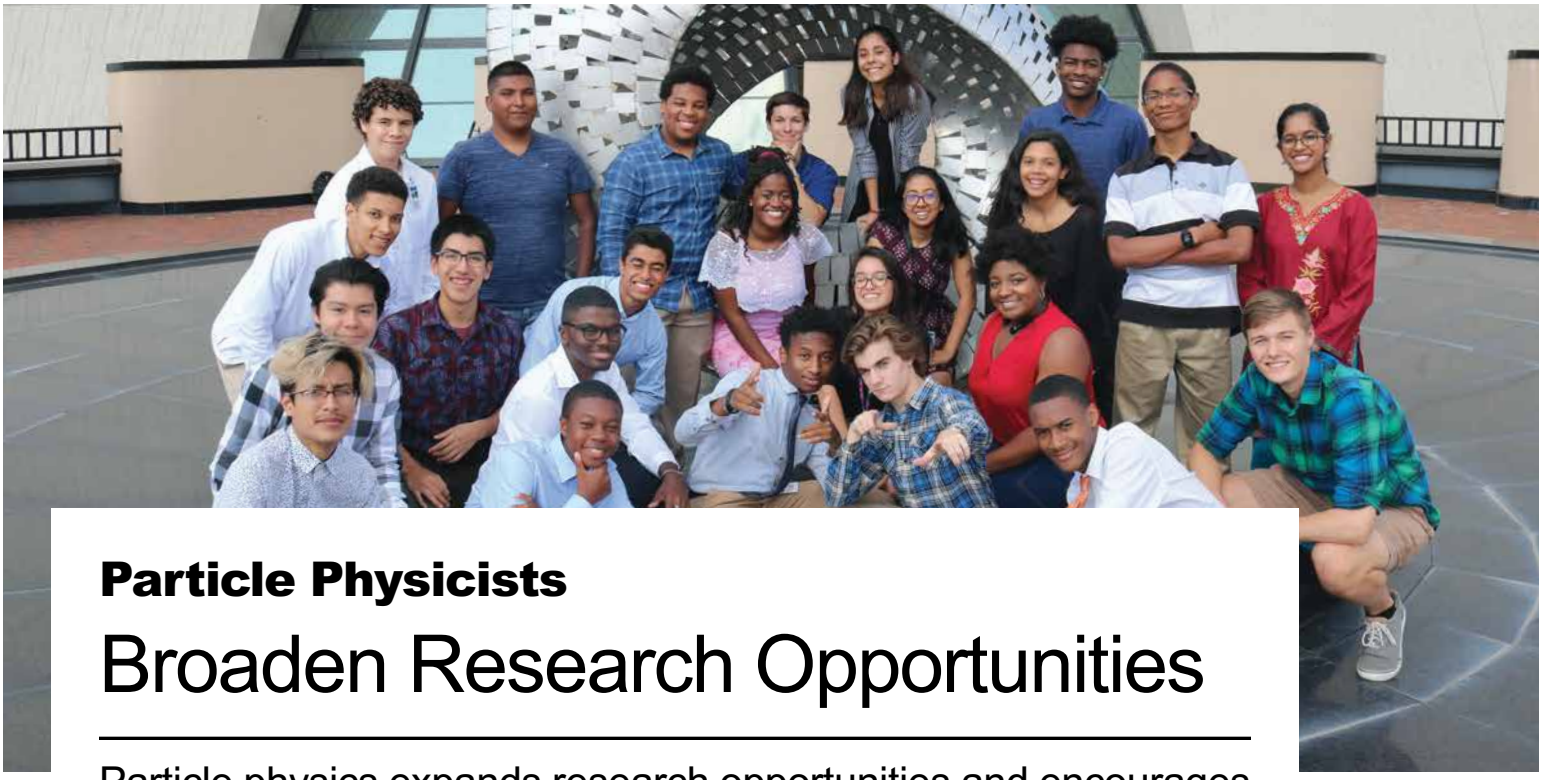
Developing a workforce

We develop our students' analytical and technical skills, enabling them to excel in all industries.

Their contributions spur innovation in technology, energy, medicine, artificial intelligence, and quantum computing.

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Particle Physicists

Broaden Research Opportunities

Particle physics expands research opportunities and encourages students to pursue advanced degrees, contributing to a strong and dynamic scientific community.



Opening Doors for Community College Students

Robert Zill worked on sensors for the IceCube Neutrino Observatory in the South Pole and connected with a global community of researchers – all as a **student from College of DuPage, a public community college** in Illinois.

Immersive Internships for Exceptional Undergraduates

*"I got to collaborate with different teams... I'm proud of myself because I finished something, and now I **can see myself finishing school and not giving up**."*

Jordan Glover, Tougaloo College, USCMS PURSUE Program Student

*"My physics professor recruited me for a [NSF] Research Experience for Undergraduates program. The adventure of a real summer job in research **inspired me to pursue science as a career**."*

Michelle Stancari, Senior Scientist, Fermilab



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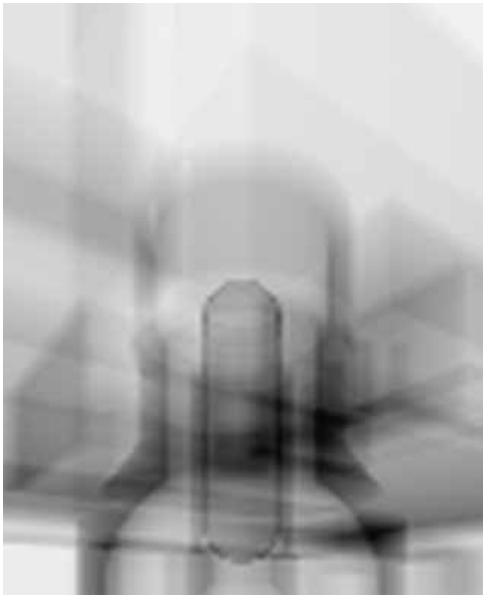
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Particle Physicists

Bolster National Security

Particle Physics plays a crucial role in national security, with physicists collaborating closely with the U.S. government to leverage their expertise and enhance the safety of our nation.



Advanced Cargo Screening

Particle physics detector technology improves homeland security by enabling advanced cargo screening.

Investigations Inside Nuclear Reactors

U.S. scientists helped create a particle detector system to safely look inside the nuclear reactors in Fukushima after the earthquake and tsunami in Japan.

Remote Monitoring of Nuclear Reactors

The PROSPECT, Chandler, EOS, and MAD neutrino detectors in the U.S. are demonstrating technology for the monitoring of nuclear reactors, supporting nonproliferation efforts.

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Particle Physicists

Advance Artificial Intelligence

As physicists, we push the boundaries of AI by using cutting-edge techniques to:

- Operate complex particle accelerators
- Enhance rapid image processing
- Minimize computational resources
- Train algorithms to filter unreliable information
- Collaborate with industry leaders
- Expand ultrafast data processing



Dr. Nhan Tran (Fermilab) develops high-speed computing techniques for ultrafast AI.

Train an AI-adept workforce



*"AI is important because it can help you discover things your brain and other techniques can't… Industry needs this expertise, and I've already worked with **Boeing, Lockheed Martin**, and companies in the **energy sector**."*

Sandra Biedron, University of New Mexico and Element Aero

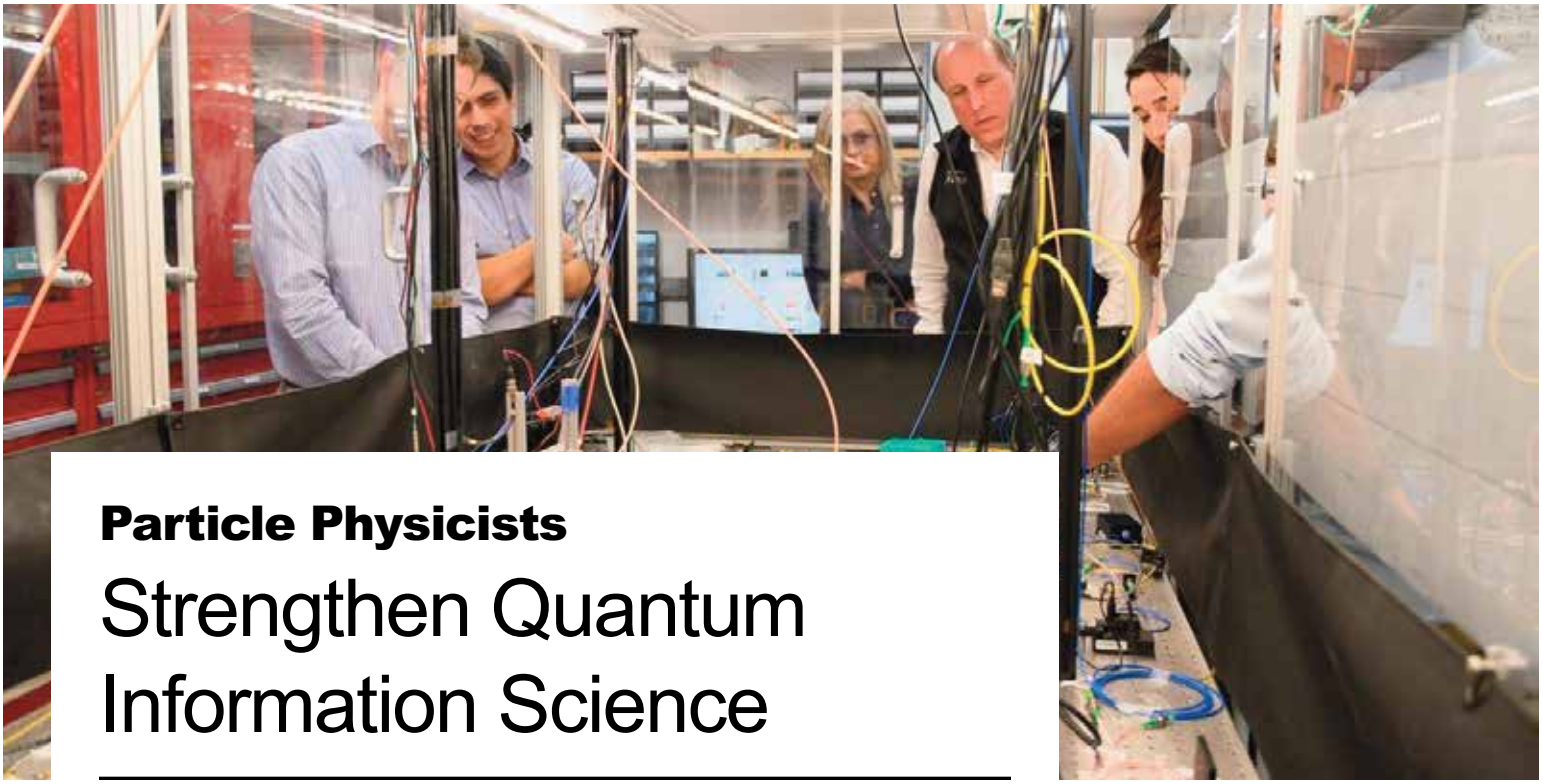
*"At **Tesla**, I transformed enormous data sets into detailed road maps for self-driving cars and at **DeepMind**, I am building AI systems that imagine and plan. My experience in particle physics trained me to tackle these complex problems while dealing with the noisy and uncertain data of the real world."*



Alex Mott, DeepMind

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Particle Physicists

Strengthen Quantum Information Science

Quantum Information Science (QIS) combines quantum mechanics and information theory to study information processing, analysis, and transmission.

World-class Research

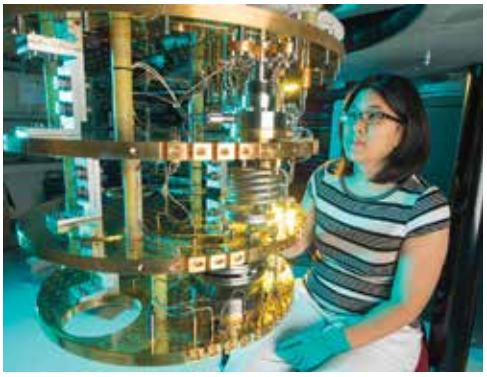
SLAC National Accelerator Laboratory is pursuing breakthroughs in quantum information science and technology with SLAC's new 5,500 sq. ft. Superconducting Quantum Foundry at its Detector Microfabrication Facility.

Work with Industry Leaders

Research in particle physics helps pave the way for breakthroughs in quantum computing, ultra-secure communications via quantum networks, and more powerful computing techniques.



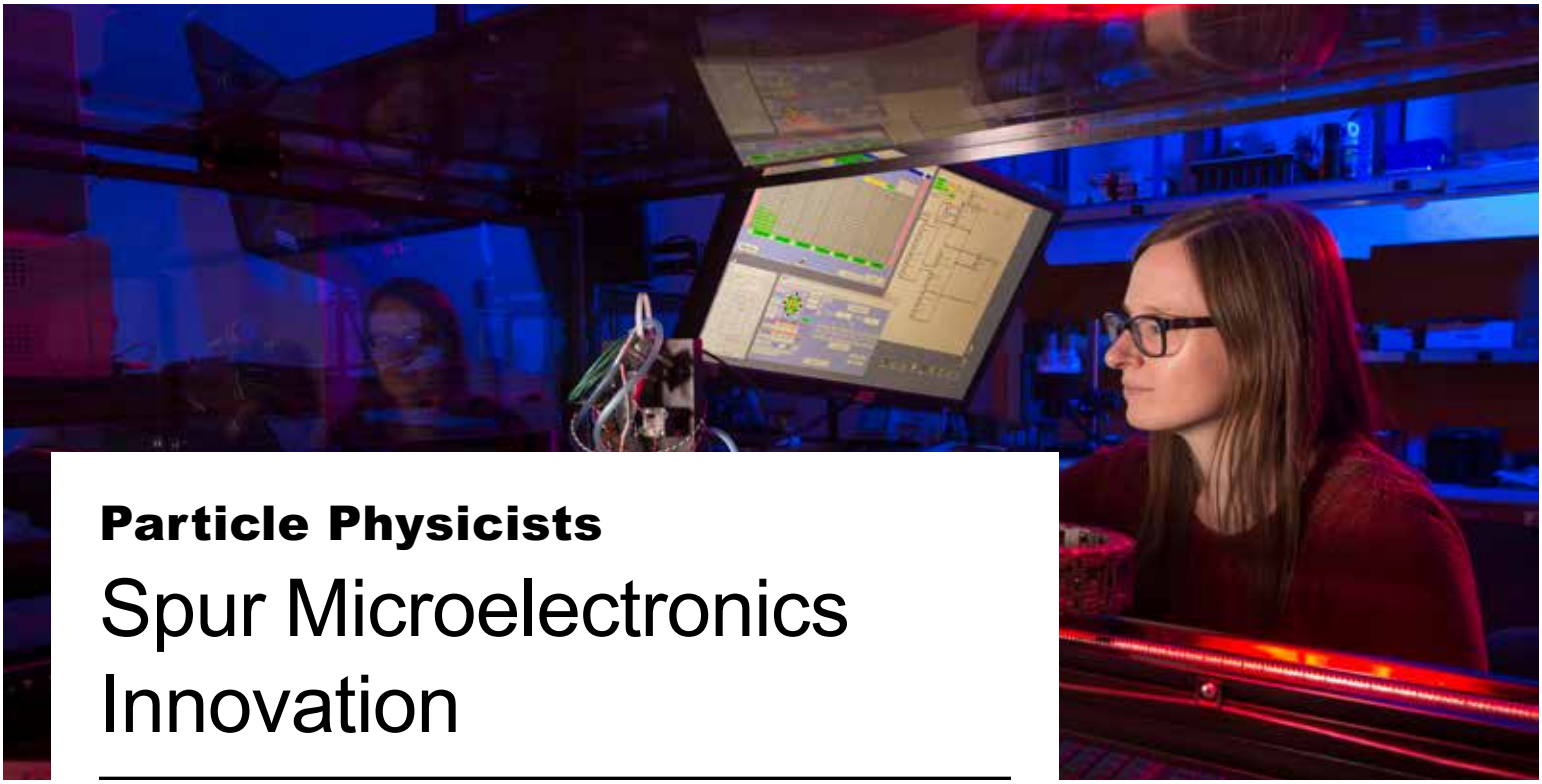
Caltech graduate student Samantha Davis analyzes quantum teleportation data.



Fermilab physicist Lauren Hsu works on a dilution refrigerator for quantum sensors.

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Particle Physicists

Spur Microelectronics Innovation

Experimental particle physics is advancing our understanding of the universe while creating technologies with real-world impacts across fields like communication, computing, and weather science.

Working with National Labs

National lab particle-beam test facilities are essential for designing and testing radiation-tolerant chips for particle physics, satellites, and space travel.

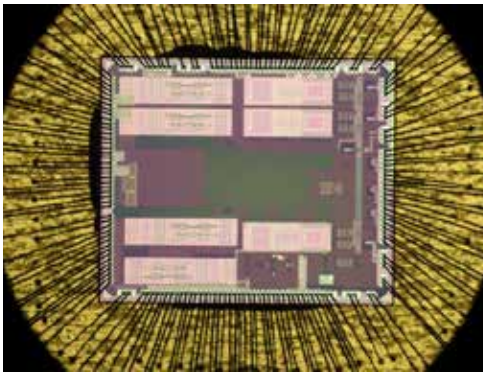
Develop a skilled workforce

The particle physics community trains students in **industry-standard tools** and workflows.

Microelectronics developed for particle physics have direct applications to avionics, space travel, medicine, and the military.



Andrew Williams at his Fermilab internship searching for dark matter



The COLUTA analog-to-digital converter chip, developed by HEP for high speed data transmission in extreme radiation environments.

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